

Evaluating an AI Documentation Assistant for Anesthesiology Teams

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Abstract

In addition to ensuring patient safety during anesthetic inductions, anesthesiologists must document clinical interventions and administer drugs. This is a time-consuming and low priority task, which harms the documentation quality of anesthetic protocols. In this case study, we demonstrate how speech-based artificial intelligence (AI) assistants that leverage closed-loop communication can increase documentation quality. An evaluation in 40 scenarios in a medical high-fidelity simulator indicated that the AI documentation assistant facilitated earlier data entry and increased documentation precision. However, despite the objective advantages for data quality and patient safety, anesthesiologists experienced a higher temporal demand with the system. With this study, we contribute

qualitative insights of how the AI documentation assistant benefited anesthesiologists' work style and affected their interactions within the team. Future research should aim to design AI assistants that enforce communication clarity while considering their impact on team dynamics.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; *Natural language interfaces.*

Keywords

Healthcare, Anesthesiology, Spoken Language Understanding, Automation, Human AI Teaming

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1 Introduction

Documentation of clinical interventions and drug administration is important for medico-legal reasons [26, 30, 33], billing and quality improvement [9], and future real-time artificial intelligence (AI) decision support systems [28]. For example, during anesthetic inductions, anesthesiologists are required to document performed interventions, such as the start of the induction, applied instrumentation, and administered drugs. However, documentation is a time-consuming task [32], and anesthesiologists' motivation for documentation is mostly driven by legal obligation [16]. As a result, anesthesia protocols meet legal requirements (e.g., the administration of a sleep-inducing drug is documented before the administration of a muscle relaxant that paralyses the patient), but this inaccurate data is less useful for quality improvement and real-time AI decision support systems (e.g., no exact administration time or real-time documentation) [3, 16]. Automation of documentation seems like an obvious solution for increasing documentation quality while alleviating the mental workload of anesthesia teams. Approaches to documentation automation include the registration of vital sign parameters [24], which is limited to the sensor data collected from the patient [2], and speech recognition, which required calling out predefined command words [1]. However, although previous systems have increased data quality, users did not experience speech recognition as more beneficial than the standard documentation [1]. In summary, there is a need for accurate documentation systems that do not need to be explicitly spoken to.

Grundgeiger et al. [16] argued that user experience and anesthesiologists' motivation should be considered to embed the documentation task more fluently in the workflow. They suggested sourcing information for automatic documentation by extracting it from the anesthesia team's existing communications. An existing type of communication that is essential for documentation is so-called *closed-loop communication*. In short, a communication loop starts with an anesthesiologist announcing the medication or interventions that should soon be administered (e.g., "Give 0.25 mg of Fentanyl"). The nurse then repeats what they heard (e.g., "0.25 mg of Fentanyl") and prepares for the administration by picking up the syringe. If no correction is announced, the nurse then administers the drug and reads back to close the loop (e.g., "0.25 mg of Fentanyl given"). Such closed-loop communication enhances patient safety because everyone in the room knows the patient's current medication status and the team has a shared situation awareness [6]. Patient safety is a main motive that underlies anesthesiologists' activities [16]. Thus, a system that enforces closed-loop communication has significant potential to improve patient safety, as an observational study showed that less than half of the communication loops initiated by either a nurse or an anesthesiologist were completely closed [15]. Since such an AI system does not yet exist, we need to understand how it would affect interactions within anesthesia teams.

In the present case study, we briefly describe the actual implementation of an AI-based speech recognition system that continuously listens to the audio stream of an anesthesia team and derives documentation intentions from it (i.e., information that the anesthesiologists are required to document in the electronic anesthesia protocol). We evaluated this AI documentation assistant with 20

anesthesia teams (i.e., anesthesiologist and anesthetic nurses) in 40 high-fidelity simulations of anesthesia induction scenarios. We (1) assessed its effectiveness as precondition of acceptance [34] and, more importantly, (2) yield insight into how anesthesia teams interacted with and perceived the AI documentation assistant.

Based on previous findings by Grundgeiger et al. [16], we expected documentation quality to increase with use of the AI documentation assistant. More specifically, we expected that (1) information would be entered with a shorter documentation time delay (i.e., the time between the intervention and its entry into the electronic anesthesia protocol), (2) information would be entered with a lower documentation time error (i.e., the absolute difference between the actual intervention time and the documented intervention time). In addition, we expected that (3) anesthesiologists' subjective mental workload would decrease with the AI documentation assistant.

Our main contributions are qualitative insights on how anesthesia teams interacted with and perceived the AI documentation assistant. We analyzed our qualitative observations and the anesthesia teams' statements to learn how the system affected interactions within teams, verbal communication, and the overall documentation process. We discuss how a prototypical AI documentation assistant that played a team support role [13] increased the documentation quality of anesthesia teams and altered their experiences and social dynamics. Finally, we contribute two challenges to be considered for the design of AI team members for safety critical domains: (1) its smooth integration into an existing team and (2) prioritizing clear communication.

2 AI Documentation Assistant

2.1 Design Process

The AI documentation assistant was designed as part of the BMBF research project CASSANDRA in which we aimed to develop an assistive system for anesthesia teams. We followed a Contextual Design process [21]. First, we observed 24 anesthetic inductions and four extubations with 25 anesthesia teams, consisting of at least one anesthesiologist and one nurse. Through four online interviews, we filled knowledge gaps regarding emergency procedures that were not observed. During interpretation sessions, observations and statements were recorded in individual notes and illustrated as models. Second, we clustered these 1200 notes bottom up to form an affinity diagram, as well as identity models, collaboration models, artifact models, sequence models, and a physical model [21]. Third, an interdisciplinary team ($n = 13$) of seven HCI researchers, three data scientists, two computer scientists, and one anesthesiologist performed two wall walks on these models and diagrams [21], which resulted in 65 initial key insights and 66 larger design ideas; these were pursued and complemented during visioning. Fourth, as part of a visioning process, we conceptualized various assistive systems, one of which was the AI documentation assistant described in this work. To lend more weight to the users' voices in choosing their prospective artificial teammate, we also conducted workshops with eight anesthesiologists and eight nurses to determine the personality exhibited by the intelligent assistive system [23]. These insights were considered in the visioning process. We iterated emerging prototypes twice in formative tests involving a total of $n = 11$ anesthesiologists.

2.2 System Features

The AI documentation assistant's features included continuous listening to communication and visual feedback on what it understood. Visual feedback of recognized drugs and interventions was logged on a tablet interface that resembled a chat. Like reading confirmations in a chat, drugs were labeled with double check marks when the nurse confirmed the administration and closed the loop (see Figure 1). Missing information, such as the dosage for an advised drug, was flagged with a question mark. When the missing information was comprehensible from the readback upon loop closure, the question mark was replaced with a value (e.g., the administered dosage). Users could scroll the protocol history by swiping up and down on the tablet.

In formative tests, we asked anesthesiologists to simulate a standard induction including communication with a nurse. Through Wizard of Oz [11] operation of the prototype, anesthesiologists could experience the system's reactions in real-time. Originally, we had started the prototyping with a plethora of features, then identified negligible functions through user tests. The removed features included automatic recognition of people in the room and adding them as "attending and responsible" into the protocol, displaying information on augmented reality glasses instead of a tablet, and touch interactions with the tablet beyond scrolling, such as editing information flagged with a question mark. We restrained the AI documentation assistant to remain passive and omitted cognitive aid features that were only appreciated by some users, such as displaying time elapsed since the administration of a drug or emitting an alert if called out medication or procedures contradicted information in the patient file. After two prototype iterations, we implemented the system.

2.3 Technical Implementation

The speech-based AI documentation assistant comprised four main components and was implemented as a pipeline [25] of AI models: (1) voice activity detection (VAD), (2) speech to text (STT), (3) intent and slot recognition (ISR) and (4) dialogue state tracking (DST). The VAD segments both audio streams individually into contiguous speech segments. STT transcribes each speech segment and ISR annotates each transcript. While VAD, STT and ISR are stateless the DST accumulates the outputs of both prior pipeline branches and updates the dialogue state and visualization in real-time. The STT model was pre-trained on $5 \cdot 10^6$ labeled speech-transcript pairs for three weeks on five NVIDIA A100 GPUs and later fine-tuned on 10^3 domain-specific samples. The STT training data was collected from multiple speakers. In addition, the ISR model was pre-trained on general text and adapted to classify intents and slots based on the STT output. The ISR training data was derived from several workshops between data scientists and anesthesiologists. The DST was implemented as a rule-based system to detect closed-loop communication and to eliminate repeated utterances. Furthermore, the DST groups related information, such as the anesthesia team's call outs after tracheal intubation. The components were implemented in Python and Rust and use the PyTorch Machine Learning library. Our models build on the Conformer architecture [19], the BERT

architecture [12] and Whisper [29]. The AI system was designed to operate fully on-premise, following an agile AI-development life-cycle that included several model refinements and user feedback. Tests of various microphone systems indicated that the best acoustic quality and least crosstalk was achieved when using consumer headsets rather than ambient- or stage microphones.

3 Method

3.1 Participants

For our case study, we recruited 20 ad hoc anesthesia teams (an anesthesiologist and a nurse) from the University Hospital Würzburg, Germany. Twenty anesthesiologists and 19 nurses with a mean age of 32 years ($SD = 6$) and eight years ($SD = 5$) of job experience participated in our case study. To complete the odd-numbered team, one anesthesiologist participated twice, and acted as a nurse in one simulation. Ethical clearance was obtained from the ethics committee of the local university, and all participants provided written informed consent.

3.2 Setup

We prepared the simulation room at the University Hospital Würzburg, Germany to replicate an actual induction room of a hospital. The main difference between the simulation and a regular induction room is that the patient bench was occupied by a human patient simulator [8] that was operated from the control room next door. Otherwise, the hardware in the simulation room was identical to that of operational induction rooms (Figure 2). The AI documentation assistant was locally deployed on a laptop in the same room. Audio information was picked up from anesthesiologists and nurses via two Bluetooth headsets that could be worn without blocking the acoustic meatus (Openrun, AfterShokz, Leiden, Netherlands). The use of these microphones and this recording setup was also approved for use in the actual operation room environment by the hospital. The AI documentation assistant's outputs were displayed in the feedback interface (Figure 1b) on a tablet (Apple iPad Air fifth generation; Figure 2), which allowed anesthesia teams to correct misinterpretations [34]. Automated transfer of the information captured by the AI documentation assistant to the hospital's proprietary patient data management system was not possible. As a workaround Author CD (anesthesiologist) manually transferred the recognized events to the anesthesia protocol using a different computer in the adjacent control room of the simulation center. This Wizard of Oz procedure [11] led to a 30 s delay due to a combination of the human wizard's reaction time and synchronization of the electronic anesthesia protocol. In accordance with hospital policies, any automatically entered data, including vital signs, needed to be acknowledged and confirmed by the anesthesiologist. Therefore, we entered the data as "prescribed" in the anesthesia record, and the anesthesiologists had to sign off on the entry for eventual documentation in the record. Using four room cameras, we audio- and video recorded the scenarios and logged participants' voice interactions, as transcribed in real-time by the AI documentation assistant.

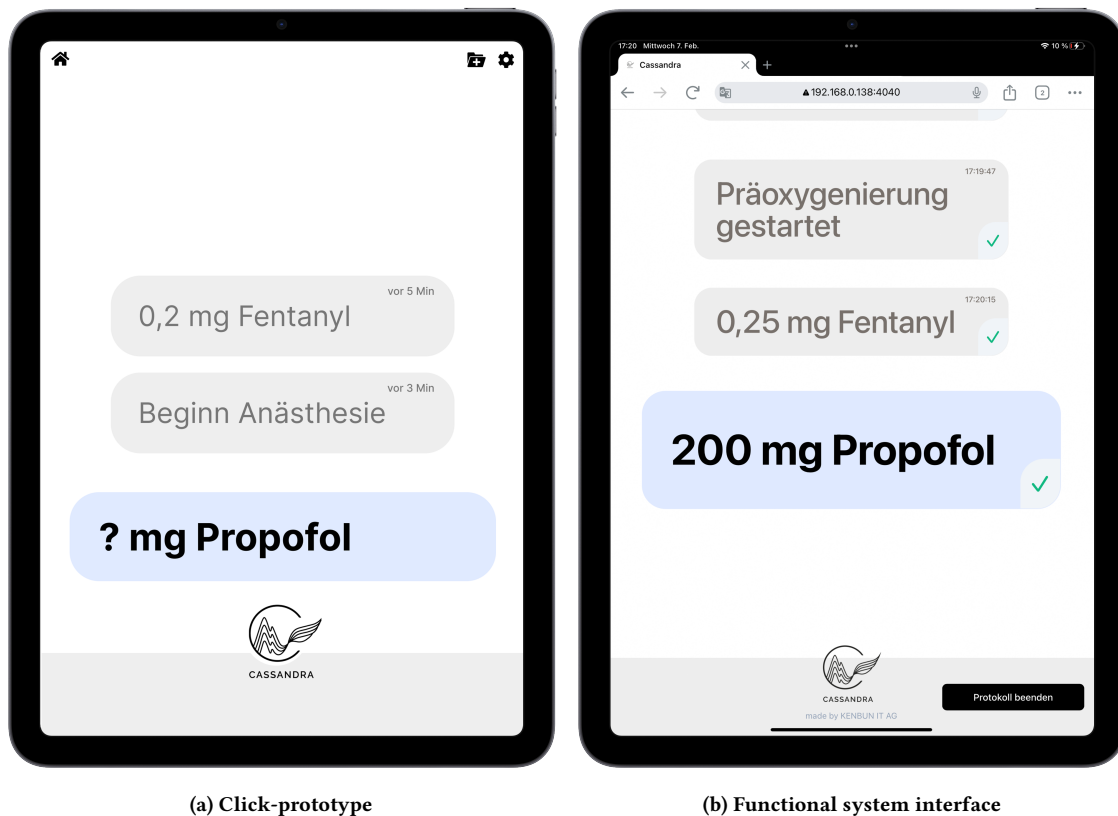


Figure 1: Screenshots of the feedback interface on the initial prototype and the running system, which informed the anesthesia teams of successful automatic documentation. If the closed-loop communication was complete, the system documented (1) at what time the anesthesiologists ordered (2) what dosage for (3) which medication and (4) whether this was confirmed by nurses as administered. Some information such as the beginning of pre-oxygenation was usually stated but did not require readback. Therefore, it was directly written into the protocol (“Präoxygenierung gestartet”). If the system registered a drug but did not understand its dosage, this was indicated via a question mark (“? mg Propofol”).

3.3 Procedure

First, participants received a short introduction to the setup of the simulation room. Second, we demonstrated the capabilities of the AI documentation assistant and prompted them to test what would happen if they, for example, announced the administration of “0.25 ml of fentanyl.” This trial also served as a final audio test for the technical setup.

Third, the teams conducted the two simulated anesthetic inductions. Each induction entailed (1) greeting the patient, checking their identity, sobriety and allergies; (2) checking the equipment and medication; (3) preoxygenation; (4) administering analgesic, hypnotic, and relaxation medication; (5) directing laryngoscopy and intubating the patient; and (6) connecting the patient to the mechanical ventilator. The first scenario was clinically uneventful, while the second scenario included a hypotension (i.e., abnormally reduced blood pressure) that resulted in an increasing heart rate after intubation. When anesthesiologists noticed the changes in both vital parameters, they would administer Akrinor® (cafedrine and theodrenaline) to counter the hypotension. Each scenario ended after the patient was stable and cleared for surgery. Note that the hypotension in the second scenario is a common occurrence, not

an emergency, and was not perceived as particularly stressful by participants.

After the second scenario, both team members completed a questionnaire about their demographic data and the NASA Task Load Index (NASA TLX, [7]) to report their perceived task load for documentation. In addition, anesthesiologists completed the NASA TLX to retrospectively state their usual documentation task load during real inductions. Finally, we interviewed both participants for 15 minutes and asked them to reflect on their experience and perceptions of interactions within the team and with the AI documentation assistant, and about suggestions for improvement.

4 Results

Overall, the AI documentation assistant recognized 1326 events. These included initial safety questions (e.g., patient identity, allergies, sobriety, and intended surgery), the announcement or confirmation of drugs (e.g., hypnotic, analgesic, relaxant, and antibiotic), verbalized interventions and checks related to tracheal intubation (e.g., difficulty, type of tube, depth, fixation, and ventilation), and preparations for surgery (e.g., catheters and eye ointment and coverage).



Figure 2: Re-staged setup showing (A) infusion pumps, (B) patient monitoring, (C) respirator, (D) patient simulator, (E) the interfaces for conventional documentation, and (F) the AI documentation assistant. The medication trolley was placed behind the anesthesiologist and is not shown in the picture. The working system can be seen in the online appendix or at the following link: <https://www.youtube.com/watch?v=CWDd88uhQHI>

4.1 Drug Documentation Quality

Although the AI documentation assistant documented a variety of events, we focused on drug documentation for the quantitative analysis, because it is the most important, best-studied [1, 3, 16], and best-communicated [15] intervention. To measure documentation time delay and error, we annotated drug administration in the video, noted recognized events from the AI documentation assistant's log files, and exported the documented administration time and the time of data edits from the anesthesia protocol. For the AI documentation assistant, we defined documentation time delay as the time between the intervention and its appearance on the tablet. Because anesthesiologists had to confirm or correct AI-documented drugs in the anesthesia protocol, we used the time stamps from the electronic anesthesia record of these manual confirmation or edits to calculate a baseline for the documentation time delay analyses (assumed time of manual documentation). For this assumed time of manual documentation, we defined documentation time delay as the time between the intervention and the event confirmation or editing off the event in the anesthesia protocol. Finally, we defined documentation precision as the absolute difference between the actual intervention time and the documented intervention time.

In all scenarios combined, we annotated a total of 212 drug administrations in the videos. The AI documentation assistant captured 191 drug events, which resulted in an apparent recognition rate

of 90% (i.e., 90% of all drugs were displayed on the AI documentation assistant's visual interface). However, some teams verbally repeated confirmations specifically for the correct capture of speech recognition. Therefore, the AI documentation assistant's technical recognition rate was lower in reality. Although anesthesiologists are obliged to review and complete automatically entered data in the anesthesia protocol, the anesthesiologists in our case study visually inspected 90 drug events in the protocol and only manually confirmed, edited, or entered 122 drug events. Out of these 122 drug events, 29 had been recognized by the AI documentation assistant but not transferred in time by the human wizard.

4.1.1 Drug Documentation Time Delay. The AI documentation assistant detected and showed drugs with a median delay of 4 s on the tablet ($IQR = 10$ s, $n = 122$). The assumed time of manual documentation had a median delay of 382 s ($IQR = 202.25$, $n = 122$). A Wilcoxon test showed this documentation time delay difference of 99% was statistically significant ($z = 9.55$, $p < .001$, $r = .996$).

4.1.2 Drug Documentation Time Error. Because the automatic timestamps were set almost immediately, documentation time delay and error were identical for the AI documentation assistant. The AI documentation assistant ($Mdn = 6$ s, $IQR = 7$, $n = 29$) showed a significantly lower documentation time error by over 92% than the assumed time of manual documentation ($Mdn = 79$ s, $IQR = 63$, $n = 29$), $z = 4.70$, $p < .001$, $r = 1.0$.

4.2 Subjective Workload

Contrary to our expectations, a paired samples t-test indicated no significant difference between the NASA TLX raw score with the AI documentation assistant in our case study ($M = 6.18, SD = 2.55$) and retrospectively generated NASA TLX raw scores based on actual inductions without documentation support ($M = 6.03, SD = 2.44$), $t(19) = .22, p = .83$. The only sub-scale that indicated a difference between these scores was temporal demands, which anesthesiologists perceived as higher when using the AI documentation assistant ($M = 7.95, SD = 3.83$) than in retrospectively rated actual inductions ($M = 5.65, SD = 3.07$), $t(19) = 2.26, p = .035, d = .51$.

4.3 Qualitative Insights

Based on observations during the induction scenarios and subsequent interviews with the teams, Authors SH and RF made qualitative notes. The notes were clustered bottom up to an affinity diagram, from which we extracted 14 insights. For the present paper, we summarized these insights into the following five overarching themes.

4.3.1 Initial expectation management. At the beginning of the first scenario, the teams were already surprised by the AI documentation assistant's abilities. That is, the safety questions on sobriety, dental state, medication, previous surgeries, and scheduled surgery were very reliably recognized. The following standard medications (i.e., fentanyl, propofol, and rocuronium) was usually also captured by the system. The teams' initial positive experiences with the AI documentation assistant raised their expectations and led them to challenge its abilities. Several teams also started to dictate information for which timeliness of documentation to the induction protocol is less essential, such as the application of thermal blankets, positioning, or body temperature. For example, one participant dictated "Remifentanyl has been running for five minutes on 0.1 mg/kg/min" (Anesthesiologist [A]11), while other teams attempted to use the AI documentation assistant to document previous interventions made by the colleague who had attended the patient before the handover (e.g., "intravenous catheter [already] in place"; A18, A8). Such retrospective documentation via voice command was not the intended use of the AI documentation assistant and was not registered by the system. Additionally, some participants wanted the AI documentation assistant to understand the team's natural communications (i.e., to be tolerant to slang and imperfect loop closure) – a system behavior we had intentionally avoided to increase communication clarity.

4.3.2 Work style adaptation for enhanced communication and patient safety. The anesthesiologists and nurses were willing to adapt their working style to increase the AI documentation assistant's automatic recognition rate. They spoke more clearly, tried to normally pronounce often omitted details (such as units), and announced medications by their chemical name rather than a brand. Team members noted and appreciated that the presence of the AI documentation assistant changed their pronunciation and fostered the practice of closed-loop communication. For example, A8 stated that, "through the direct feedback, you check more rigorously what you are doing, which improves patient safety," while A14 commented

that, "with [the AI documentation assistant], I communicated more consciously, as if explaining it to someone who is doing this for the first time and not someone who has been doing this for a thousand times." Particularly for documentation related to the infusion pump, anesthesia teams were not accustomed to vocalizing all units in perfusors, (e.g., "Fentanyl runs on 6 mg/kg/hour"). Still, several teams took on this challenge and repeatedly uttered the confirmation until the AI documentation assistant visually notified them that the drug had been understood.

4.3.3 (No) need to closely monitor the system, as every piece of automatically documented information helps. Most anesthesiologists sought confirmation about whether the AI documentation assistant had understood them and frequently focused on the screen. Indeed, criticism that some anesthesiologists raised is that their urge for feedback captured their visual attention. Varying latencies between utterance and recognition (approximately 1 – 4 s) contributed to this issue. Other anesthesiologists were unbothered by the presence of the AI documentation assistant and even began private conversations, as they would in actual inductions once the patient is asleep. These anesthesiologists were unconcerned about visual feedback for their utterances and consequently did not pay attention to the tablet. They appreciated some level of automatic documentation in the protocol but did not aim to increase the AI documentation assistant's recognition rate or even repeat not recognized terms. For example, A19 stated, "I do not need live visual feedback. I will see later what worked and what did not. [...] Some medication was not recognized. It is too cumbersome to repeat that again and again." Moreover, A3 commented, "If [the AI documentation assistant] does not recognize my 'is administered,' I would not repeat it for [the system]."

However, anesthesiologists acknowledged the potential of the system or even praised the increased quality in documentation data. They found already the automatic real-time documentation of only few events supportive and used these as "anchors" to document other interventions that were not automatically registered. For this, they either used confirmed drugs and interventions from the protocol or searched timestamps of medication that had been announced but not logged as confirmed on the tablet (i.e., the loop had not been closed).

4.3.4 Partially filled in data in the anesthesia protocol interrupted the documentation work flow. In our case study, overall documentation efficiency likely did not increase due to the AI documentation assistant's recognition of only some interventions and drugs. Anesthesiologists were still required to complete the entire protocol and even felt that their usual workflow and click-paths were interrupted because some of the form fields already contained data. While the number of clicks and key-strokes may have decreased, their cognitive mental effort did not. For instance, one anesthesiologist found that "[the AI documentation assistant] recognized the peripheral venous catheter but not the details (that is, 16 gauge, right arm), so the effort of adding that was higher" (A16), and A15 stated "The effort of double-checking everything in the protocol is higher than directly entering everything by myself" (A15). Anesthesiologists preferred to work with the entries recognized from their voice on

the tablet that ran the AI documentation assistant rather than the desktop documentation interface for the anesthesia protocol.

4.3.5 Automatic documentation depends on the nurse, but the anesthesiologist is responsible. Overall, anesthesiologists considered the AI documentation assistant as helpful and potentially very useful in dynamic situations or for complex induction procedures followed by short surgeries. In the latter case, anesthesiologists do not usually have enough time to catch up with documentation while the surgery is ongoing. By contrast, nurses did not benefit from the time savings or reduced effort associated with the AI documentation assistant because documentation is the responsibility anesthesiologists' responsibility. However, nurses also did not experience any increased effort or disadvantages when participating in closed-loop communication. One nurse (N20) "attempted to give readback using the anesthesiologist's wording," while another nurse "felt encouraged by the visual feedback to close the communication loop" (N19). As loop closure was crucial for automatic recognition in our algorithm, the nurses' commitment to readback was critical for the recognition of drug administrations.

5 Discussion

The use of the AI documentation assistant improved documentation quality by reducing documentation time delay and error. Given that we simulated non-complex induction scenarios and that documentation time delay and error in actual inductions are much higher [3, 16], we expect the benefits of the AI documentation assistant to be even greater in more complex scenarios and an actual operating room environment.

Contrary to our expectations but in line with Grundgeiger et al., the AI documentation assistant did not reduce the anesthesiologists' workload [16]. Anesthesiologists even rated their temporal effort with the AI documentation assistant as higher than in retrospective workload ratings of actual inductions. One obvious possible explanation might be the setting and retrospective rating of actual inductions. Simulated and actual operating room environments have shown differences in, for example, visual attention distribution [18] and communication [27], which may have affected the present workload ratings. Second, for the AI documentation assistant to fulfil its role as a team support [13], it requires anesthetic teams to communicate in closed loops. This benefits patient safety but may not always be practiced and thus be perceived as additional temporal effort. Third, the completion of the anesthesia protocol and crosschecking of events that were automatically recorded from speech recognition changed the anesthesiologists' form-completion workflow. Knowledge workers' resistance to changing their working with raw data from scratch to simply completing or correcting the shortcomings of an AI system has been previously reported [4]. However, clinical documentation integrity specialists have learned to benefit from cooperating with imperfect automation [5].

Finally, using novel systems may be exciting (or frustrating), and users need time to learn the benefits and limitations of technology [17]. Therefore, our next step is to test the AI documentation assistant in actual induction rooms to overcome simulation artifacts and address how repeated use affects interactions with and perceptions of the system.

5.1 Challenges for Human-AI Teaming in Health HCI

In general, the AI documentation assistant was positively perceived by users; however, the qualitative data indicated possibilities for improvement. We identified two larger challenges for the development of future AI systems in healthcare that are perceived as supportive team members rather than a "second patient" [20].

5.1.1 Challenge 1: Creating Perfectionist AI that Asserts Clear Communication to Benefit Patient Safety. By design, our AI documentation assistant insisted on closed-loop communication for medication, and its understanding of spoken language was restricted to a list of predefined medical terms. The latter contained communications observed in early user research and was filtered by our medical partners to only support clear and concise communication. As we observed in actual induction scenarios, loop closure was sometimes omitted (see also [15]), partially communicated (e.g., nurses responding with a short "given" instead of reading back the drug and dosage), or non-verbally conveyed (e.g., via eye contact). Participants in our case study perceived the enforced clarity as the AI system's immaturity in mastering natural language. As communication clarity contributes to patient safety [6], AI documentation assistants should enforce it, even at the cost of an initially lower recognition rate and a more formal working atmosphere. Establishing clear communication is a considerable challenge. Even in the safety-critical model field of aviation, where international standard phraseology [14] facilitates verbatim presets in documentation interfaces [22], non-adherence to formal phraseology limits automated event detection and requires more flexible speech recognition [10].

5.1.2 Challenge 2: Avoiding the Disruption of Group Development Caused by the Introduction of an AI Team Member. Anesthesiologists and nurses sometimes team up ad hoc for an anesthetic induction without ever having met before. Following Tuckman's [31] four-stage group development model, this should initiate a process of *forming*, *storming*, and *norming* before reaching the *performing* stage. In anesthesiology, however, predefined roles and shared knowledge in highly standardized procedures enable anesthesia teams to start each induction from *norming*, the third stage of team development, and directly proceed to the *performing* portion of their tasks. Introducing the AI documentation assistant interrupted this pattern because it redistributed the documentation effort from being the anesthesiologist's sole responsibility to a dependence on the nurse's readback quality. This requires orientation efforts that are typical for *forming*, the first stage in Tuckman's model [31]. To alleviate any disruptions caused by the reallocation of the documentation effort and maintain nurses' motivation, we should more carefully cater to the nurses' needs in future iterations. Therefore, the second challenge is for researchers to equally include all team members instead of solely prioritizing the person whose task is directly supported by the AI assistant as the introduction of a new AI team member will affect the entire team. This disruption even increases, when patients are active users of the AI system. Briefing patients on the purpose and skills of an AI assistant may bind valuable time then lacking for healthcare provision [34].

6 Conclusion

In this case study, we contribute qualitative insights on how the introduction of an AI documentation assistant affected working style and team dynamics in simulated anesthetic inductions. Although good system performance is a prerequisite for increased documentation quality, the social team role of AI assistants must be carefully designed to enable them to assert themselves over issues such as communication clarity and ultimately improve patient safety.

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